

Take Test: Midterm

Test Information

Description

Instructions

Timed Test This test has a time limit of 4 hours. You will be notified when time expires, and you may continue or submit.
Warnings appear when **half the time, 5 minutes, 1 minute, and 30 seconds** remain.

Multiple Attempts This test allows 8 attempts. This is attempt number 1.

Force Completion This test can be saved and resumed later. The timer will continue to run if you leave the test.
Your answers are saved automatically.

QUESTION 1

0.5 points

Save Answer

A dimensional model can contain more than one fact table.

- ☐ True
- ☐ False

QUESTION 2

0.5 points

Save Answer

The command **SELECT REPLACE('mnopq', 'o','x') FROM DUAL;** returns which of the following values?

- ☐ A. mnopqox
- ☐ B. oxoox
- ☐ C. MNOPQox
- ☐ D. mnxpq

QUESTION 3**0.5 points**

Save Answer

Bitmap indexes tend to use less space than a _____ because they use bits instead of bytes to store their data.

- ☐ a. hash index
- ☐ b. sparse index
- ☐ c. B-tree index
- ☐ d. reverse index

QUESTION 4**0.5 points**

Save Answer

Which of the following statements about how a company would use AWS Organizations are accurate? (Select TWO.)

- ☐ A. A company can only manage AWS Organizations through the AWS Management Console.
- ☐ B. A company can use AWS Organizations to create security groups that control access to resource.
- ☐ C. A company can consolidate and centrally manage multiple AWS accounts.
- ☐ D. A company can use AWS Organizations' consolidated identity and access management (IAM) feature to replace the existing IAM system for an individual account.
- ☐ E. A company can benefit from volume discounts from consolidated billing.

QUESTION 5**0.5 points**

Save Answer

Statement A: The Query Parser optimizes and simplifies a query and then passes it on to the Query Executor.

Statement B: In the DBMS architecture, the Storage Manager takes care of concurrency control.

Which sentence(s) is/are correct?

- ☐ Only A
- ☐ Only B
- ☐ A and B
- ☐ Neither A or B

QUESTION 6**0.5 points**

Save Answer

Which of the following is *not* true about indexes?

- ☐ a.

They can improve performance when MySQL searches for rows in a table.

- ☐ b. They can be deleted by using the DROP INDEX statement.
- ☐ c. They can help speed up insert, update and delete operations on columns that are updated frequently.
- ☐ d. MySQL automatically creates an index for unique constraints.

QUESTION 7

0.5 points

Save Answer

A(n) ____ contains a copy of a primary key value from a parent table.

- ☐ clone key
- ☐ foreign key
- ☐ constraint
- ☐ index

Remaining Time: 3 hours, 55 minutes, 08 seconds.

✖ Question Completion Status:

SELLER

<u>SellerID</u>	SellerName	RealtorID
S111	Paul	R1
S222	Carly	R1
S333	Ana	R2
S444	Charlotte	R2
S555	Aidan	R3
C666	Eric	R1
C777	Christina	R1

REALTOR

<u>RealtorID</u>	RealtorName
R1	Claire
R2	Kate
R3	Cliff
R4	William

This database will be used for the following questions citing tables from the HOMETOWN REALESTATE database:

If a DBMS enforces a DELETE RESTRICT option on the referential integrity constraint between SELLER and REALTOR in the HOMETOWN REALESTATE database, what will be the outcome

after a user tries to delete the first record (S111, Paul, R1) from SELLER?

- ☐ SELLER will have 7 records, REALTOR will have 4 records
- ☐ SELLER will have 6 records, REALTOR will have 3 records
- ☐ SELLER will have 7 records, REALTOR will have 3 records
- ☐ SELLER will have 6 records, REALTOR will have 4 records
- ☐ SELLER will have 3 records, REALTOR will have 3 records

QUESTION 9

0.5 points

Save Answer

Within one column, one row of a relation can contain two values from that column's domain.

☐ True

Click Save and Submit to save and submit. Click Save All Answers to save all answers.

QUESTION 10

0.5 points

Save Answer

RAID systems use a single disk to create storage volumes.

- ☐ True
- ☐ False

QUESTION 11

0.5 points

Save Answer

The maximum cardinality indicates the maximum number of relationship instances in which an entity participates.

- ☐ True
- ☐ False

QUESTION 12

0.5 points

Save Answer

All of the following are record-based models *except*:

- ☐ A. network.
 - ☐ B. hierarchical.
 - ☐ C. entity-relationship.
 - ☐ D. relational.
-

QUESTION 13**0.5 points**

Save Answer

Which of the following describes the completeness data quality characteristic?

- ☐ The extent to which data correctly reflects the real-world instances it is supposed to depict
 - ☐ The degree to which all the required data is present in the data collection
 - ☐ The characteristic that requires each real-world instance to be represented only once in the data collection
 - ☐ The extent to which the data properly conforms to and matches up with the other data
 - ☐ The degree to which the data is aligned with the proper time window in its representation of the real-world
-

QUESTION 14**0.5 points**

Save Answer

In both SQL and Oracle, user-defined structured types are called:

- ☐ A. classes.
 - ☐ B. entities.
 - ☐ C. object types.
 - ☐ D. attribute data types.
-

QUESTION 15**0.5 points**

Save Answer

The hierarchical data model was developed in the _____.

- ☐ a. 1960s
 - ☐ b. 1970s
 - ☐ c. 1980s
 - ☐ d. 1990s
-

QUESTION 16**0.5 points**

Save Answer

Autonumber data type option:

- ☐ Automatically converts the existing data type into a number type
 - ☐ Assigns a number to each of the used data types
 - ☐ Enables automatic counting of columns in a table
 - ☐ Enables automatic counting of tables in a database
 - ☐ Enables automatic generation of consecutive numeric data values in a column
-

QUESTION 17**0.5 points**

Save Answer

The online version of the MySQL Reference Manual lets you

- ☐ a. use the links in the left sidebar to drill down to the information you're looking for
- ☐ b. search for a specific word or phrase
- ☐ c. return to the Home page
- ☐ d. all of the above

QUESTION 18**0.5 points**

Save Answer

Every attribute, atomic or composite, plays one of three roles in a relation schema. It is a key attribute, a non-key attribute, or a candidate key.

- ☐ True
- ☐ False

QUESTION 19**0.5 points**

Save Answer

The following query returns a single number as a total (or sum) of all blocks, used by all datafiles in the database: ____.

- ☐ SELECT TOTAL (BLOCKS) FROM DBA_DATA_FILES;
- ☐ SELECT SUM (BLOCKS) FROM DBA_DATA_FILES;
- ☐ SELECT AVG (BLOCKS) FROM DBA_DATA_FILES;
- ☐ SELECT COUNT (BLOCKS) FROM DBA_DATA_FILES;

QUESTION 20**0.5 points**

Save Answer

_____ and its console can be used as a centralized control center for many of the available DBA tools.

- ☐ Enterprise Manager
- ☐ Buffer Cache
- ☐ Process Monitor
- ☐ Job Queue Process

QUESTION 21**0.5 points**

Save Answer

What is Amazon Q Developer?

- ☐ a. A wizard that leads you through the steps to configure your integrated development environment (IDE)
- ☐ b.

A generative AI-powered coding assistant

- ☐ c. A browser-based CLI
- ☐ d. A cloud-based integrated development environment (IDE)

QUESTION 22

0.5 points

Save Answer

Data redundancy removes the potential for inconsistencies in the stored data.

- ☐ True
- ☐ False

QUESTION 23

0.5 points

Save Answer

In creating methods for a user-defined type in Oracle, those that are created for the type for use without an implicit parameter are called:

- ☐ A. dynamic methods.
- ☐ B. static methods.
- ☐ C. procedures.
- ☐ D. functions.

QUESTION 24

0.5 points

Save Answer

The attribute hierarchy provides a top-down data organization that is used for two main purposes: _____ and drill-down/roll-up data analysis.

- ☐ a. decomposition
- ☐ b. de-normalization
- ☐ c. normalization
- ☐ d. aggregation

QUESTION 25

0.5 points

Save Answer

Renaming a redo log file may be done to keep up with new naming standards or simply to make the filenames more meaningful to you.

- ☐ True
 - ☐ False
-

QUESTION 26**0.5 points**

Save Answer

A block of code is a self-contained sequence of one or more commands.

- ☐ True
 - ☐ False
-

QUESTION 27**0.5 points**

Save Answer

Assume a relationship requires a construct that involves three superclasses and a subclass in a single relationship type. This relationship can be modeled as a(n)_____.

- ☐ categorization
 - ☐ aggregation
 - ☐ specialization/generalization
 - ☐ logical schema
-

QUESTION 28**0.5 points**

Save Answer

Entity clustering as an ER modeling construct is a useful abstraction to present an ER diagram at a narrower level of conceptualization.

- ☐ True
 - ☐ False
-

QUESTION 29**0.5 points**

Save Answer

In the EER model, if entity instances cannot belong to more than one subclass, we say the subclasses are:

- ☐ A. total.
 - ☐ B. partial.
 - ☐ C. complete.
 - ☐ D. disjoint.
-

QUESTION 30**0.5 points**

Save Answer

When associated with the LIKE operator, _____ represents a *series of one or more unspecified characters*.

- ☐ %
- ☐ _
- ☐ &
- ☐ ^

QUESTION 31**0.5 points**

Save Answer

In mapping a relational model to an ER model, if a relation has a composite primary key and each part of the composite key is a foreign key, the corresponding ER construct is:

- ☐ A. a relationship.
- ☐ B. a strong entity with a multi-valued attribute.
- ☐ C. a strong entity.
- ☐ D. a strong entity with a weak entity.

QUESTION 32**0.5 points**

Save Answer

ROLAP and MOLAP vendors are working toward the integration of their respective solutions within a unified decision support framework.

- ☐ True
- ☐ False

QUESTION 33**0.5 points**

Save Answer

Stored procedures must have at least one argument.

- ☐ True
- ☐ False

QUESTION 34**0.5 points**

Save Answer

In the notational scheme used in the chapter for the Presentation layer ER model, both the cardinality ratio and the participation constraint are expressed via a "look across" approach.

- ☐ True
- ☐ False

QUESTION 35**0.5 points**

Save Answer

The first transaction timestamp for an optimistic concurrency control technique is assigned at what time?

- ☐ A. At the end of the validation phase
- ☐ B. At the start of the write phase
- ☐ C. At the start of the transaction
- ☐ D. At the end of the read phase

QUESTION 36**0.5 points**

Save Answer

_____ is the process of rendering the undesirable FDs desirable.

- ☐ Leveling
- ☐ Normalization
- ☐ Clearing
- ☐ Clarification

QUESTION 37**0.5 points**

Save Answer

In the optimistic approach, during the _____ phase, changes are permanently applied to the database.

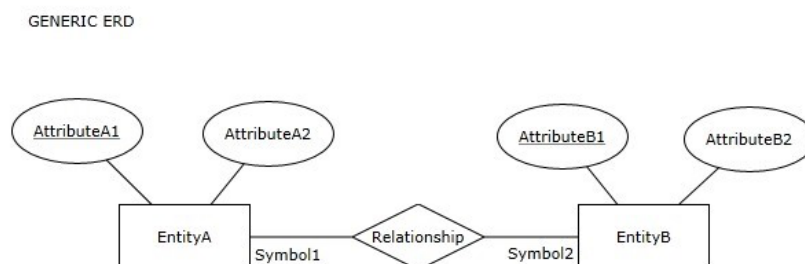
- ☐ a. validation
- ☐ b. shared
- ☐ c. read
- ☐ d. write

QUESTION 38**0.5 points**

Save Answer

GENERIC ERD

Observe the GENERIC ER Diagram below:



If in the GENERIC ERD the Symbol 1 is $>|$ and the Symbol2 is $0<$, then the number of relations in the relational schema resulting from mapping the GENERIC ERD is 2.

- ☐ True
- ☐ False

QUESTION 39**0.5 points**

Save Answer

SQL*Plus accepts standard SQL, with all the Oracle enhancements to SQL, such as the TO_DATE function.

- ☐ True

☐ False

QUESTION 40

0.5 points

Save Answer

Master data management's main goal is to provide a partial and segmented definition of all data within an organization.

- ☐ True
- ☐ False

QUESTION 41

0.5 points

Save Answer

Direct path loads appending new blocks is the best option when rows are deleted from tables because existing blocks will never have rows slotted into them.

- ☐ True
- ☐ False

QUESTION 42

0.5 points

Save Answer

If the column named float_value is a DOUBLE type, which of the following queries will *not* return values that are approximately equal to 5?

- ☐ a. **SELECT float_value**
FROM float_sample
WHERE float_value = 5
- ☐ b. **SELECT float_value**
FROM float_sample
WHERE ROUND(float_value, 2) = 5.00
- ☐ c. **SELECT float_value**
FROM float_sample
WHERE float_value BETWEEN 4.99 AND 5.01

QUESTION 43

0.5 points

Save Answer

___ is an integrity constraint applicable only to SQL-updatable views.

- ☐ LOCAL
- ☐ CHECK OPTION
- ☐ WITH CASCADED
- ☐ WITH LOCAL

QUESTION 44**0.5 points**[Save Answer](#)

The Administration screen of the Database Control provides comprehensive maintenance access to database utilities, backup/recovery, and deployment activities.

- ☐ True
 - ☐ False
-

QUESTION 45**0.5 points**[Save Answer](#)

Archived data from source systems is never present in the data warehouse.

- ☐ True
 - ☐ False
-

QUESTION 46**0.5 points**[Save Answer](#)

The scope of a scalar variable is limited to

- ☐ a. the script in which it's declared
 - ☐ b. the cursor in which it's declared
 - ☐ c. the block in which it's declared
 - ☐ d. the statement in which it's declared
-

QUESTION 47**0.5 points**[Save Answer](#)

Which statement is CORRECT?

- ☐ The OID of an object remains the same during the entire lifetime of the object.
 - ☐ An OID is the same as a primary key in a relational database setting.
 - ☐ Two objects with the same values always have the same OID.
 - ☐ Each literal is defined by an OID according to the ODMG standard.
-

QUESTION 48**0.5 points**[Save Answer](#)

The Oracle isolation level that guarantees transaction-level consistency for all transactions is:

- ☐ A. read only.

- ☐ B. read committed.
- ☐ C. serializable.
- ☐ D. write only.

QUESTION 49

0.5 points

Save Answer

The information stored in the _____ is used by the DBMS for a recovery requirement triggered by a ROLLBACK statement, a program's abnormal termination, or a system failure such as a network discrepancy or a disk crash.

- ☐ a. data dictionary
- ☐ b. metadata
- ☐ c. transaction log
- ☐ d. rollback manager

QUESTION 50

0.5 points

Save Answer

How many entities are involved in a unary relationship?

- ☐ 1
- ☐ 2
- ☐ More than 2
- ☐ Between 1 and 2
- ☐ Less than 1

QUESTION 51

0.5 points

Save Answer

DOCTOR

Observe the table DOCTOR:

DOCTOR

<u>DoctorID</u>	Name	DateOfMDGraduation	NumberOfPatients
D111	John	1/1/1999	100
D222	Tina	1/2/2003	117
D333	Ellen	1/1/2001	117
D444	Tina	1/1/2003	144

This table will be used for the following questions citing table DOCTOR.

Which of the following functional dependencies does NOT exist in the table DOCTOR?

- ☐ DoctorID --> Name
- ☐ DoctorID --> DateOfMDGraduation
- ☐ DoctorID --> NumberOfPatients
- ☐ Name --> DoctorID
- ☐

All of the above exist in the table DOCTOR

QUESTION 52**0.5 points**[Save Answer](#)

The ER model is classified as:

- ☐ A. semantic.
- ☐ B. external.
- ☐ C. seminal.
- ☐ D. hierarchical.

QUESTION 53**0.5 points**[Save Answer](#)

An ORDBMS will typically support inheritance...

- ☐ only at a tuple level
- ☐ only at data type level
- ☐ only at table type level
- ☐ both at data type and table type level

QUESTION 54**0.5 points**[Save Answer](#)

All of the following statements about forward engineering are true *except*:

- ☐ A. the source model is typically more expressive than the target model.
- ☐ B. the target model is typically more expressive than the source model.
- ☐ C. the source model is closer to the mini-world being modeled than the target model.
- ☐ D. the target model is usually more efficient for implementation than the source model.

QUESTION 55**0.5 points**[Save Answer](#)

A distributed database is composed of several parts known as database _____.

- ☐ a. parts
 - ☐ b. partitions
 - ☐ c. sections
 - ☐ d. fragments
-

QUESTION 56**0.5 points**

Save Answer

Database _____ means protecting the database from unauthorized access, modification, or destruction.

- ☐ A. privacy
- ☐ B. security
- ☐ C. integrity
- ☐ D. confidentiality

QUESTION 57**0.5 points**

Save Answer

Hadoop is a database that has become the de facto standard for most Big Data storage and processing.

- ☐ True
- ☐ False

QUESTION 58**0.5 points**

Save Answer

True or False? In the relational model, each row is called a tuple.

- ☐ True
- ☐ False

QUESTION 59**0.5 points**

Save Answer

A common use of a trigger is to create a historical record of a transaction.

- ☐ True
- ☐ False

QUESTION 60**0.5 points**

Save Answer

Which statement is CORRECT?

- ☐ One of the main disadvantages of an OODBMS is that persistence orthogonality is not supported.
- ☐ An OODBMS guarantees the ACID properties.
- ☐ An OID is the same thing as a primary key.
- ☐ Two objects are identical or equivalent when the values of their variables are the same.